Capstone Project: Multi-Agent AI Platform with GenAI, Analytics & Azure Deployment

Left Shift Program 2026 – Data & AI (T5)
Type: Individual Project
Complexity: Intermediate → Expert

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1. Project Overview

Students will design, build, and deploy an **end-to-end Multi-Agent AI Platform** that integrates:

- **Python Full-stack development**
- **Machine Learning / Deep Learning**
- **GenAI (LLMs, RAG, embeddings, multi-agent orchestration)**
- **Azure AI services**
- **Azure Data Engineering & Analytics stack**
- **Power BI visualization**
- **Azure deployment / CI-CD fundamentals**

The project simulates a real-world Data + AI engineering scenario, requiring the student to integrate skills learned across the full track.

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2. Problem Statement (Choose ONE of the following domains)

Students must pick one domain and build their end-to-end solution around it:

1. **Smart Retail Assistant** – Demand forecasting + customer Q&A + anomaly detection
2. **Intelligent Healthcare Support System** – Symptom triage + medical information assistant + patient data analytics
3. **Financial Risk & Advisory Platform** – Fraud detection + portfolio insights + regulatory search assistant
4. **Manufacturing Quality & Productivity Suite** – Defect classification + production optimization + maintenance insights
5. **Custom domain scenario (with SME consent)**

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3. Mandatory Components

A. Python Fullstack (Backend + APIs)

Students must create:

- Python-based backend using **Flask or FastAPI**

- Minimum **4 REST APIs** (data ingestion, ML prediction, document search, agent interaction)
- Integration with any one database:
    - **SQL** (Azure SQL) or
    - **NoSQL** (MongoDB, CosmosDB)
- Basic **logging, error handling, and unit testing (pytest)**

B. Machine Learning / Deep Learning

Implement at least **one ML/DL model**, e.g.:

- Regression / classification (sklearn)
- Clustering
- Time-series forecasting
- Image classification (CNN / OpenCV)

Deliverables:

- Clean data pipeline
- Feature engineering
- Training + evaluation
- Model persistence (pickle / ONNX)

C. GenAI / Agents

Build a **minimum 2–3 agent system** using:

- Prompt engineering
- Embeddings + vector store
- Retrieval-Augmented Generation (RAG)
- Agents (Autogen / LangChain / CrewAI / Azure GenAI)

Examples:

- **Data Analyst Agent** – answers questions from analytics data
- **Document Assistant Agent –** searches PDFs / knowledge base
- **ML Expert Agent –** helps generate insights from ML outputs

Implement:

- Multi-agent orchestration

- Agent communication through Model Context Protocol (MCP) *optional but recommended*

D. Azure AI & Cloud

Use at least **three Azure components**:

- **Azure OpenAI / Azure AI Studio**
- **Azure Bot Service** or Web App
- **Azure Cognitive Services** (Search, Vision, Document Intelligence, Text Analytics)
- **Azure ML** for training or inference
- **Azure AI Foundry** for deployment patterns

Include:

- Deployment diagram
- Security consideration (Key Vault, environment variables)

E. Data Engineering Pipeline

Using any combination of:

- **Azure Data Factory –** ingest raw data
- **Azure Databricks (PySpark) –** clean, transform, analyze data
- **Azure Fabric –** Lakehouse, pipelines, Data Activator
- **SQL** (Spark SQL or T-SQL)

Deliver:

- Raw → Staged → Curated data flow
- Delta tables or parquet-based storage

F. Analytics & Visualization

Build a **Power BI dashboard** showing:

- Key metrics
- Model outputs
- Anomaly alerts / trends
- Agent-driven insights (optional)
- Publish & share report

G. Final Deployment

Deploy the solution on Azure using one of:

- **Azure Web App / Container App**
- **Docker + GitHub Actions CI/CD**
- **Azure App Service**
- **Serverless Functions (for tasks)**

Your multi-agent system must be **live / executable** during evaluation.

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4. Deliverables

Deliverable	Description
Technical Documentation	Architecture, data flow, models, prompts, agents, APIs
Working Code Repository	GitHub link with clear folder structure
Deployment Diagram	Azure components + interactions
Configuration Files	Environment, pipeline definitions, YAML, etc.
Power BI Report	Final analytics dashboard
Demo Video (5–10 mins)	Walkthrough of platform
Reflection Note	Challenges + learnings + optimizations

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5. Evaluation Criteria (50% Weightage)

Area	Weight
Design & Architecture	10%
Python Full stack & APIs	10%
ML/DL Model Quality	8%
GenAI & Multi-Agent Implementation	12%
Azure Deployment & Data Engineering	8%
Documentation & Presentation	2%

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6. Optional Bonus Features (+5% Extra Credit)

- MCP-based agent interoperability
- Real-time streaming analytics (Event Hub + Data Activator)
- LLM fine-tuning or custom embeddings
- Automated CI/CD with GitHub Actions